

MINA ABDEL-MASIH

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Education

North Carolina State University

May 2028

Bachelor of Science in Computer Science (AI Concentration), Math Minor | GPA: 4.0/4.0

Raleigh, NC

- **Ranked #1 in Computer Science (306), Engineering (2,205), and Junior Class (7,793)**
- **Honors/Awards:** Henry B Rowe Jr. Scholar, Robert F. & Romaine S. Angel Scholar, Dean's List – Fall 2024, Spring 2025, Fall 2025, Spring 2026
- **Courses:** Data Structures & Algorithms, Machine Learning & Data Analysis, C & Software Tools, Software Development

Technical Skills

Programming Languages (Proficiency): Java, C, Python, LaTeX, HTML/CSS, JavaScript

Technologies: Git, GitHub, Microsoft Office, VSCode, Eclipse, Google Workspace, React.js, Claude Code, Postman, N8N, Jenkins

Certifications: ChatGPT Prompt Engineering for Developers, IBM Unleashing The Power of AI Agents

Work Experience

Information Security Research | North Carolina State University

Aug. 2024 – Dec. 2024

Personal Research

Raleigh, NC

- Investigated privacy risks and mitigation strategies across 3 levels (individual, organizational, societal); synthesized insights from 6 peer-reviewed studies (2009–2021) on authentication (2FA/biometrics), encryption, and human-factor drivers (privacy fatigue, awareness, common mistakes), highlighting evidence that enabling 2FA reduces compromise risk by $\approx 100\%$
- Engineered a **full-stack research showcase website** using **HTML, CSS, and JavaScript**, achieving 100% responsive design and smooth interactive transitions; leveraged AI-assisted development tools to accelerate design and debugging, demonstrating the ability to translate complex research into a fast, accessible product

Zaki Technology

May 2026 – Present

Automation Developer (Contract)

Raleigh, NC

- Scoped an **AI workflow automation** for an enterprise client to replace a manual 2–3 hour daily data-entry process, analyzing the existing pipeline and architecting an email-ingestion, classification, and human-review system using **N8N**, the **Gemini API**, and **Google Workspace**
- Designed a **human-in-the-loop MVP** with a structured **JSON** classification schema (loss type, status, and confidence score), prioritizing reliability and review safeguards before any automated database writes
- Engineered an automated **email-parsing and labeling workflow** to ingest 20–30 daily alerts, extracting article titles, URLs, and snippets for downstream AI classification and structured **Google Sheets** output

Projects

Access Control Compliance Engine | Java, JUnit, CSV I/O, Custom Data Structures

February 2026

- * Designed and built a **Java-based compliance tool** for an electronic medical records client, automatically parsing user activity logs against access control rule sets to classify events as violations, warnings, or compliant actions across datasets of 262,000+ records
- * Implemented **custom list-based Map and sorting algorithms from scratch** to enable multi-key sorted report generation (by username, timestamp, action, and resource), meeting strict sub-33 second performance benchmarks on large-scale inputs
- * Delivered the **full software development lifecycle** end-to-end: authored a formal proposal, produced a UML class diagram, wrote a structured system test plan, implemented the solution in Java, and validated correctness through **JUnit testing**

Student Productivity Tool | Java 21, Spring Boot, JUnit 5, React, REST API, FullCalendar

May 2026

- * Architected a **full-stack student productivity web app** using **Java 21 and Spring Boot**, exposing **15+ REST API endpoints** for full CRUD operations across courses, assignments, and lecture notes with custom business logic validators and sentinel value patterns
- * Engineered a **modular Java backend** with **JUnit 5 test coverage** across all calculator classes handling weighted grade computation, GPA calculation, course status classification (ON_TRACK, AT_RISK, CRITICAL), and deadline detection
- * Built a **responsive React frontend** with a reusable component library, dynamic dashboard with live grade statistics, tabbed course detail pages, a **FullCalendar-integrated deadline view**, and a complete notes system with full CRUD functionality

Rooted | IBM x NCCU AI Hackathon | watsonx, LangChain, FastAPI, Python, PostgreSQL, Docker, Redis, Streamlit

March 2026

- * Built an **agentic AI therapy journaling app** at the IBM x NCCU Hackathon, serving as the **AI/ML lead** responsible for architecting and validating all **IBM watsonx** model behavior across safety classification, crisis routing, **NLP sentiment scoring**, and contextual follow-up generation using **LangChain** and **IBM watsonx Orchestrate**
- * Engineered a **three-tier LLM fallback strategy** (watsonx REST API $\rightarrow$ LangChain ReAct agent with **granite-13b-chat-v2** $\rightarrow$ deterministic pipeline) with full **AgentTrace observability**, ensuring zero-downtime and correct model outputs under failure conditions
- * Built a **crisis detection pathway** with zero LLM latency and a **role-based async REST API** (Python, FastAPI, PostgreSQL, asyncpg, SQLAlchemy, JWT); containerized via **Docker Compose** with health checks and auto-seeded demo data, reducing setup to **under 2 minutes**

MinaAbdel-Masih.com | Portfolio Website | React, Vite, JavaScript, CSS

August 2026

- * Rebuilt a personal portfolio as a component-based **React + Vite** single-page app, architecting **10+ modular, reusable components** with data-driven content files and per-component CSS for a maintainable, cleanly separated codebase
- * Engineered interactive UI features from scratch — an **HTML Canvas** animated starfield background, **IntersectionObserver** scroll-spy navigation, frosted-glass section theming, and a client-side FAQ chat widget — integrating a **Google Apps Script** contact form and deploying to **Vercel** with a custom domain and **100% responsive design**

Student Organizations & Extracurriculars

Hackpack | Artificial Intelligence Club | NCSU App Dev Club | NCSU Oriental Orthodox Club (*Treasurer*) | UNC Coptic Student Organization | NCSU Arab Student Association | UNC Arab Student Association | NCSU Society of Sales Engineers